(Task 1 of 8) In this tutorial, we will review previously learned content. 0

Python | 1

(Task 2 of 8) What is the result of running this program?

Python [Run ▶️]	Scala 3
`m = [ 82, 76 ]`	`val m = Buffer[Any](82, 76)`
`m[0] = m`	`m(0) = m`
`print(m[1])`	`println(m(1))`

```
76
```
2

You predicted the output correctly 🎉🎉🎉 3

`m` is bound to a array. `m[0] = m` replaces the `0`-th element of the array with the array itself. This is fine because a array element can be any value, including itself. Besides, the array is not copied, so the mutation finishes immediately.

Click [here](#) to run this program in the Stacker.

Python | 4

(Task 3 of 8) What is the result of running this program?

Python [Run ▶️]	Scala 3
`n = m * 4`	`val n = m * 4`
`m = 3`	`val m = 3`
`print(n)`	`println(n)`
`print(m)`	`println(m)`

```
12 3
```
5

The answer is `error`. 6

Textual explanation

You might think `m * 4` is able to refer to `m` and hence evaluate to a value. However, `m` has not been bound to a value when `m * 4` is evaluated. In SMoL, every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

Click [here](#) to run this program in the Stacker.

Python | 7

What is the result of running this program?

What is the result of running this program?

Python [Run ▶️]

```
baz = bar + 1
bar = 2
print(baz)
print(bar)
```

JavaScript

```
let baz = bar + 1;
let bar = 2;
console.log(baz);
console.log(bar);
```

`error` 8

You predicted the output correctly 🎉🎉🎉 9

Python | 🎲 10

(Task 4 of 8) What is the result of running this program?

Python [Run ▶️]

```
v1 = [ 23 ]
v2 = v1
v1[0] = 45
print(v2)
```

Pseudo

```
let v1 = vec[23]
let v2 = v1
v1[0] = 45
print(v2)
```

`[23]` 11

The answer is `[45]`. 12

Textual explanation

You might think `v1` and `v2` refer to different arrays, so changing `v1` doesn't change `v2`. However, they refer to the same array. In SMoL, a array can be referred to by more than one variable, and binding a array to a new variable does not create a copy of that array.

Click [here](#) to run this program in the Stacker.

Python | 🎲 13

What is the result of running this program?

Python [Run ▶️]

```
a = [ 37, 26 ]
b = a
a[0] = 87
print(b)
```

Pseudo

```
let a = vec[37, 26]
let b = a
a[0] = 87
print(b)
```

`[87,26]` 14

You predicted the output correctly 🎉🎉🎉 15

Python | 🎲

(Task 5 of 8) What is the result of running this program? 16

Python [Run ▶️]

```
n = 5
def f1(m):
    def f2():
        l = 4
        return n + m + l
    return f2()
print(f1(1) + 3)
```

Lispy

```
(defvar n 5)
(deffun (f1 m)
  (deffun (f2)
    (defvar l 4)
    (+ n m l))
  (f2))
(+ (f1 1) 3)
```

`13` 17

You predicted the output correctly 🎉🎉🎉 18

This program binds `n` to `5` and `f1` to a function, and then evaluates `f1(1) + 3`. The value of `f1(1) + 3` is the value of `f2() + 3`, which is the value of `(n + m + l) + 3`, which is the value of `(5 + 1 + 4) + 3`, which is `13`.

Click [here](#) to run this program in the Stacker.

Python | 🎲

(Task 6 of 8) What is the result of running this program? 19

Python [Run ▶️]

```
m1 = [ 77, 77 ]
def f(m2):
    m2[0] = 43
    return m2[0]
foo = f(m1)
print(m1)
```

Lispy

```
(defvar m1 (mvec 77 77))
(deffun (f m2)
  (vec-set! m2 0 43)
  (vec-ref m2 0))
(defvar foo (f m1))
m1
```

`[43, 77]` 20

You predicted the output correctly 🎉🎉🎉 21

The program first creates a two-element array. The elements are both `77`. After that, the program defines a function `f`. The function call `f(m1)` replaces the first array element with 43. After that, the array is printed. The array now refers 43 and 77, so the result is `[43, 77]`.

Click here to run this program in the Stacker.

Python | 22

(Task 7 of 8) What is the result of running this program?

Python [Run ▶️]

```
def foobar(n):
    m = 4
    return n + m
print(foobar(2) + m)
```

Pseudo

```
fun foobar(n):
  let m = 4
  return n + m
end
print(foobar(2) + m)
```

```
error
```

23

You predicted the output correctly 🎉🎉🎉 24

`foobar(2) + m` is evaluated in the top-level block, where `m` is not defined. So, this program errors.

Click here to run this program in the Stacker.

Python | 25

(Task 8 of 8) What is the result of running this program?

Python [Run ▶️]

```
v1 = [ 53 ]
v2 = [ 72, v1 ]
v1[0] = 72
print(v2)
```

Lispy

```
(defvar v1 (mvec 53))
(defvar v2 (mvec 72 v1))
(vec-set! v1 0 72)
v2
```

```
[72, [72]]
```

26

You predicted the output correctly 🎉🎉🎉 27

`v1` is bound to a one-element array. The only element of this array is 53. `v2` is bound to a two-element array. The elements of this array are 72 and the one-element array. The subsequent `v1[0] = 72` mutates the one-element array. Finally, the value of `v2` is printed.

Click here to run this program in the Stacker.

You have finished this tutorial 🎉🎉🎉

Please print the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

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